

Po-Han Chen

✉ pohan@stanford.edu | 🌐 <https://pohantw.github.io>

Education

Stanford University

Stanford, CA

Ph.D. in Electrical Engineering

2021–2026

Advisor: Priyanka Raina

Thesis Title: *Designing CGRAs for Sparse ML Acceleration and Scalable Physical Implementation*

National Tsing Hua University (NTHU)

Hsinchu, Taiwan

M.S. of Electrical Engineering

2016–2018

B.S. of Electrical Engineering and Computer Science

2012–2016

Chip Research Experiences

Zircon: Heterogeneous Accelerator SoC

Stanford, CA

- Served as SoC integrator and physical designer for a heterogeneous accelerator combining a specialized systolic array and a flexible coarse-grained reconfigurable array (CGRA)
- Integrated an ARM CPU, a global buffer, a 64×32 systolic array, and a 16×16 CGRA into a complete SoC
- Implemented the chip in Intel 16 nm technology and completed physical design, timing closure, and signoff

Opal: CGRA SoC for Sparse Machine Learning

Stanford, CA

- Led the tapeout of an Intel 16 nm CGRA SoC with primitive-level optimizations for sparse ML workloads
- Developed CGRA architectural optimizations that reduced runtime and energy by 66–79% across evaluated sparse-ML benchmarks
- Established and pipe-cleaned the first complete Intel 16 nm implementation flow in the group, spanning synthesis, physical design, timing signoff, DRC, and LVS
- Designed a custom PCB and performed post-silicon bring-up, functional validation, performance characterization, and power measurement
- Automated CGRA firmware generation and testing to enable rapid application deployment
- Presented the work at the CICC conference and the ISSCC poster sessions

Open-Source CGRA

Stanford, CA

- Taped out an 8×4 CGRA using the open-source SkyWater 130 nm process and the OpenROAD EDA tools
- Integrated OpenRAM SRAM macros and a provided harness SoC with our tool-generated CGRA array
- Demonstrated technology portability of our CGRA-generation tools across an open-source CMOS flow
- Presented the work at the ISCAS conference

SAPIENS: RRAM-Based Non-Volatile Associative Memory

Stanford, CA

- Built an FPGA-based validation platform to generate configurable test stimuli, control the chip, and verify measured outputs
- Developed an on-the-fly tuning and testing flow that enabled rapid post-silicon characterization
- Results published at the Symposium on VLSI Technology conference

Digital Refocusing Accelerator

NTHU, Hsinchu

- Developed a TSMC 40 nm ASIC for accelerating block-based digital light-field refocusing
- Designed a highly parallel microarchitecture for view synthesis, exploiting pixel-, line-, and view-level parallelism to achieve up to $350 \times$ speedup over CPU execution and enable real-time refocusing
- Contributed across RTL design, functional verification, logic synthesis, and post-silicon chip measurement
- Presented the work at the Symposium on VLSI Circuits conference

Work Experiences

SEG Pixel IP Design Team, Apple

Cupertino, CA

Hardware Designer (Intern)

2022

- Explored and implemented hardware architectures for a newly developed internal algorithm, achieving significant area reduction compared with the legacy design.

Camera ISP Pipeline Hardware Design Team, MediaTek

Hsinchu, Taiwan

Hardware Designer (Full Time)

2019–2020

- Designed and verified digital IPs for camera image signal processing (ISP) pipelines, including functional verification, constrained random tests, coverage analysis, and synthesis.
- Developed automation tools to streamline internal hardware design and verification workflows.

Selected Publications

Full list: <https://pohantw.github.io>

P.-H. Chen, B. W. Cheng, M. Oduoza, Z. Xie, R. Lu, S. G. Ravipati, K. Koul, A. Carsello, Y. Mei, M. Horowitz, and P. Raina, “Opal: A 16-nm Coarse-Grained Reconfigurable Array SoC for Full Sparse Machine Learning Applications,” *IEEE Solid-State Circuits Letters (SSC-L)*, 2025.

P.-H. Chen, B. W. Cheng, M. Oduoza, Z. Xie, K. Koul, S. G. Ravipati, Y. Mei, R. Lu, A. Carsello, M. Horowitz, and P. Raina, “Opal: A 16nm Coarse-Grained Reconfigurable Array for Full Sparse ML Applications,” *IEEE Custom Integrated Circuits Conference (CICC)*, 2025.

P.-H. Chen, C. Tsao, and P. Raina, “An Open-Source 4×8 Coarse-Grained Reconfigurable Array Using SkyWater 130 nm Technology and Agile Hardware Design Flow,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2023.

P.-H. Chen, S.-W. Yang, and C.-T. Huang, “A 250-mW 5.4G-Rendered-Pixel/s Realistic Refocusing Processor for High-Performance Five-Camera Mobile Devices,” *IEEE Open Journal of the Solid-State Circuits Society (OJ-SSCS)*, 2023.

P.-H. Chen, S.-W. Yang, S.-Y. Huang, L.-D. Chen, and C.-T. Huang, “A 250mW 5.4G-Novel-Pixel/s Photorealistic Refocusing Processor for Full-HD Five-Camera Applications,” *IEEE Symposium on VLSI Circuits (VLSI)*, 2019.

- A. Carsello, J. Thomas, A. Nayak, P.-H. Chen, M. Horowitz, P. Raina, and C. Torng, “mflowgen: A Modular Flow Generator and Ecosystem for Community-Driven Physical Design”, in *ACM/IEEE Design Automation Conference (DAC)*, 2022.

- H. Li, W.-C. Chen, A. Levy, C.-H. Wang, H. Wang, P.-H. Chen, W. Wan, et al., “SAPIENS: A 64-kb RRAM-Based Non-Volatile Associative Memory for One-Shot Learning and Inference at the Edge”, in *IEEE Transactions on Electron Devices*, vol. 68, no. 12, pp. 6637–6643, 2021.

- H. Li, W.-C. Chen, A. Levy, C.-H. Wang, H. Wang, P.-H. Chen, W. Wan, et al., “One-Shot Learning with Memory-Augmented Neural Networks Using a 64-kbit, 118 GOPS/W RRAM-Based Non-Volatile Associative Memory”, in *Symposium on VLSI Technology*, 2021.

Skills

HW/SW Languages: Verilog, SystemVerilog, C, Python, Tcl

EDA Tools & Others: Simulators (VCS, Xcelium), Linting (Spyglass, nLint), Synthesis (DC, Genus), P&R (Innovus, ICC), LEC, DRC/LVS, Verdi, Git